

Four questions this dissertation refused to outsource

left — the question; right — what the field's standard answer was structurally underspecified to recover for this problem

Is the field's autonomous stability criterion the right object to measure for a driven reservoir?

$\rho(W) < 1$ verifies a static property of the recurrent matrix; it does not characterize how the reservoir contracts under the input it actually receives. For a reservoir whose job IS the driven response, the autonomous criterion is structurally underspecified.

→ Experiment A

Can the reservoir's choice be justified from a theorem about the data, not from architectural fashion?

The Maass-Markram licensing argument is non-constructive for a specific dataset — it tells you reservoirs can separate, not whether mine does for these ERPs. Takens' theorem makes a measurable claim about the data the reservoir must accommodate; I wanted that measurement.

→ Experiment B

Can the operating hyperparameter be derived rather than tuned?

Validation-accuracy-driven selection optimizes for the metric, not for the underlying information-theoretic property. A measured memory-capacity peak provides a stable, theoretically-anchored operating point that does not move when the downstream task changes.

→ Experiment C

Do clinical effects survive the strictest possible null — channel permutation — at the per-disorder level?

Parametric p-values assume a model of the noise structure that EEG channel selection does not satisfy. A channel-permutation null assumes nothing about the channels; it tests the claim the author actually wants to make.

→ Experiment D

Each question is answered, with measurement, in the experiments that follow.